



Multi-Tenant HR Payroll System

Product Requirements Document (PRD)

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1. Executive Summary

This document outlines the complete requirements for the Multi-Tenant HR Payroll System developed by Symbiosis. The system is designed to manage employee information, attendance tracking, wage calculations, and payroll processing for multiple client organizations. The platform operates under a pre-registration model with phased rollout of client-facing features.

2. Project Overview

2.1 Scope & Objectives

- Build a comprehensive multi-tenant HR payroll management system for India-based organizations
- Manage employee data, attendance, salary structures, and wage processing
- Support statutory compliance (EPF, ESI, statutory registers)
- Provide client-specific views and employee self-service portal
- Enable auditor access for compliance verification

2.2 Project Phases

- Phase 1 (Week 1): Architecture design and skeleton development
- Phase 2: Organization and employee management modules
- Phase 3: Attendance and wage processing
- Phase 4: Auditor and compliance features

3. User Types & Access Control

The system supports four distinct user roles:

User Type	Primary Functions
ERP User	Symbiosis company staff managing overall system and client accounts
Employer/Client	Client organization administrators uploading employee data, attendance, managing payroll
Employee	View-only access to payslips, reports, and personal information downloads
Auditor	Access to statutory registers, verification documents, compliance reports. Currently out of scope

4. Organization & Establishment Management

4.1 Establishment Registration

Organizations are pre-registered by Symbiosis with initial credentials. The registration collects:

- Organization name and type (Private Ltd, Partnership, etc.)
- Complete address (4 address lines + pin code)
- Primary contact: mobile, email, point of contact name
- Employer details: name, designation, mobile, email
- Documents: signature, company stamp, logo
- Auto-generated organization ID and verification code

4.2 Profile Management

- Limited editing allowed for client users (password change only)
- Admin/ERP users can modify organization details except core identifiers

4.3 Vendor/Contractor Management

Support modular vendor company onboarding for multiple employees from same contractor:

- Separate vendor company profiles under main employer
- Same data requirements as organization registration
- Primary employer linked to all vendor employees

5. Employee Information System

5.1 Core Employee Data

The system maintains comprehensive employee records:

- Unique employee ID (auto-generated)
- Personal info: name (as per Aadhaar), father's name, DOB, gender
- Marital status, nationality, education qualification
- Employment details: department, designation, date of joining
- Statutory: Aadhaar, PAN, EPF/ESI eligibility flags
- Bank details: account holder name, account number, IFSC code
- Address: permanent (4 fields) and present address (4 fields)
- Biometric: specimen signature, thumb impression, photograph

5.2 Data Upload & Excel Template

Employer can bulk upload employees via standardized Excel template with fields matching the core data structure.

- Validation: All mandatory fields required
- Aadhaar & PAN format validation
- Bank account IFSC code validation

6. Salary Structure Configuration

Organizations configure salary calculation rules applicable to all employees:

6.1 Configuration Parameters

- Basic pay percentage of CTC (e.g., 60%)
- Allowances: Dearness Allowance (DA), House Rent Allowance (HRA), conveyance, medical
- Bonuses: regular bonus, special bonus calculations
- Provident Fund: calculation ranges (5 lakh, 10 lakh, 15 lakh brackets)
- PF contribution thresholds and slab-based rates

6.2 Salary Processing

- Single salary record per employee (non-historical in current phase)
- Automatic calculation based on organization settings
- Manual override capability for specific cases

7. Attendance Management

Attendance is the primary input for wage calculation.

7.1 Upload Process

- Monthly attendance sheet upload by employer
- 100% employee coverage requirement (all employees in sheet)
- Excel or CSV format with standardized columns

7.2 Attendance Codes

System recognizes the following codes:

Code	Description
P	Present - Full working day
WO	Weekly Off - Scheduled non-working day (typically Saturday/Sunday)
H	Holiday - National/state/organizational holiday
EL	Earned Leave - Paid leave used by employee
CO	Compensatory Off - Leave granted in lieu of working on holiday

7.3 Absent Days Calculation

- Any day not marked with above codes is counted as Absent
- Absent days trigger salary deduction
- System tracks payable days for wage calculation

7.4 Data Quality & Error Handling

- Reject uploads with missing employees
- Alert for invalid attendance codes
- Duplicate attendance handling (keep latest, archive prior)
- Late submission flagging

8. Wage Calculation & Processing

8.1 Wage Slip Components

Monthly wage slip includes:

- Days in month, working days, present days, payable days
- Basic salary, allowances (HRA, DA, conveyance, medical)
- Gross salary and deductions (PF, ESI, tax, absent deduction)
- Net payable amount
- Overtime: dates, hours, total overtime hours paid
- Payment details: date, time, place of payment

8.2 Processing Workflow

- Triggered by attendance upload
- System calculates all components based on salary config
- Generate wage statements for all employees
- Display for review before finalization
- Lock for modification once approved

8.3 Handling Revisions & Adjustments

Key constraint: Once wage processed and paid, modifications are complex.

- Additional payments tracked separately
- Corrections noted with justification
- Next month adjustment option for discrepancies

9. System Architecture

9.1 Multi-Tenancy Model

- Single application instance serving multiple client organizations
- Complete data isolation per organization
- Organization-specific settings and configuration
- Vendor/contractor companies linked to parent organization

9.2 Technology Stack

Frontend: React.js, Angular.js with modern UI/UX

Backend: Node.js (or preferred backend framework)

Database: Design to support organization-level data partitioning

Cloud Storage: Secure document storage for biometric data, certificates

10. User Interfaces & Workflows

10.1 Homepage

- Company branding and information
- Navigation links (About Us, Services, Team, Contact)
- Prominent login/access buttons for different user types

10.2 Login & Authentication

- Role-based login pages (ERP, Employer, Employee, Auditor)
- Username/password authentication
- Initial password reset on first login
- Session timeout and security measures

10.3 Employer Dashboard

- Employee data management (add, edit, bulk upload)
- Attendance upload interface
- Wage/payslip generation and review
- Organization profile management
- Reports (payroll summary, statutory registers)

10.4 Employee Portal

- View personal profile and employment details
- Download payslips (PDF format)
- View wage history and salary statements
- Download form documents (Form 12BB, etc.)

10.5 Auditor Dashboard

- Access statutory registers
- Download verification documents
- View compliance status
- Generate audit reports

11. Key Features & Modules

11.1 Excel Import/Export

- Standardized Excel templates for employee data
- Bulk employee upload with validation
- Monthly attendance sheet import
- Export payroll data and wage statements

11.2 Document Management

- Upload and store employee biometric images
- Store employer signatures and company stamps
- Manage organizational documents
- Generate printable documents (payslips, certificates)

11.3 Statutory Compliance

- Statutory registers (ESI, EPF monthly statements)
- Form 12BB generation for tax purposes
- Annual payroll compliance reports
- Form D export for labor department

11.4 Reporting

- Monthly payroll summary reports
- Employee-wise salary breakdown
- Attendance analysis and trends
- Exception reports (late joiners, absences)

12. Technical Requirements

12.1 Data Security

- Encrypted storage for sensitive data (Aadhaar, bank details)
- SSL/TLS for all data transmission

- Role-based access control (RBAC)
- Audit logs for all transactions

12.2 Performance

- Support for organizations with 1000+ employees
- Bulk attendance upload processing within minutes
- Report generation within seconds

12.3 Reliability

- 99.5% uptime commitment
- Automated backups (daily/weekly)
- Disaster recovery procedures

12.4 Compliance

- GDPR compliance for data privacy
- India labor law compliance
- Tax calculation as per current fiscal year rates
- Regular security audits

13. Development Timeline

Phase	Deliverables
Phase 1	• Architecture design complete • Website skeleton with home, about, services, team, contact pages • Dummy data integration
Phase 2	• Establishment registration module • Employee onboarding and data entry • Organization login and profile management
Phase 3	• Salary structure configuration • Attendance upload and processing • Wage calculation and payslip generation
Phase 4	• Employee login and payslip download • Auditor dashboard and compliance reports • Testing with real data from 2-3 organizations

14. Constraints & Special Considerations

14.1 Data Management Constraints

- Attendance upload triggers wage processing (cannot separate)
- Once wage processed and paid, modifications are complex
- Single salary record per employee (no historical records stored currently)
- 100% employee coverage required in attendance sheets

14.2 Business Rules

- All employees must belong to an organization
- Vendor/contractor employees linked to primary employer

- Salary configuration applies organization-wide
- Pre-registration model: Only Symbiosis creates new organizations

15. Future Enhancements (Out of Scope - Phase 1)

- Client-facing registration portal (organizations self-register)
- Historical wage records and salary revisions
- Advanced leave management (leave balance, accrual)
- Mobile app for employees
- Biometric attendance integration
- API for third-party integrations
- Advance salary and loan module
- Performance appraisal integration

16. Acceptance Criteria

The project will be considered successfully complete when:

- All four user roles can successfully authenticate and access appropriate modules
- Employee data can be uploaded and retrieved without errors
- Monthly attendance upload triggers correct wage calculations
- Payslips generated match configured salary structures
- All statutory registers generate correctly
- Employees can download payslips in PDF format
- Auditors can verify all compliance documents
- System tested successfully with real data from 2-3 organizations
- All security requirements met (encryption, access control, audit logs)

Database Schema Overview(Main Tables)

Organizations Table

org_id | org_name | org_type | address1 | address2 | address3 | pin_code | mobile | email |
employer_name | employer_designation | employer_mobile | employer_email |
employer_signature | company_stamp | company_logo | verification_code

Employees Table

emp_id | org_id | emp_name | father_name | dob | gender | marital_status | nationality |
education | department | designation | doj | aadhaar | pan | epf_eligible | esi_eligible |
bank_name | account_no | ifsc | permanent_address | present_address | signature |
thumb_impression | photo

Salary Configuration Table

config_id | org_id | basic_pay_pct | da | hra | conveyance | medical | bonus_pct |
special_bonus | pf_slab_5L | pf_slab_10L | pf_slab_15L

Attendance Table

attendance_id | org_id | emp_id | month | year | attendance_date | attendance_code |
upload_date

Wage Statement Table

wage_id | org_id | emp_id | month | year | days_in_month | working_days | present_days |
payable_days | basic | da | hra | conveyance | medical | gross | pf_deduction | esi_deduction
| tax | absent_deduction | net_payable | payment_date | payment_status

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